



INDIA : CLIMATE, VEGETATION AND WILDLIFE

You read in newspapers daily and watch on T.V. or hear others talking about weather. You must know that **weather** is about *day to day changes* in the atmosphere. It includes changes in temperature, rainfall and sunshine etc. For example, as such it may be hot or cold; sunny or cloudy; windy or calm. You must have noticed that when it is hot continued for several days you don't need any warm clothing. You also like to eat or drink cold things. In contrast there are days together, you feel cold without woollen clothes when it is very windy and chilly, you would like to have something hot to eat.

Broadly, the major seasons recognised in India are:

- Cold Weather Season (Winter) December to February
- Hot Weather Season (Summer) March to May
- Southwest Monsoon Season (Rainy) June to September
- Season of Retreating Monsoon (Autumn) October and November

COLD WEATHER SEASON OR WINTER

During the winter season, cool, dry winds blow from north to the south. The sun rays do not fall directly in the region as a result, the temperatures are quite low in northern India.

HOT WEATHER SEASON OR SUMMER

In the hot weather season sun rays more or less directly fall in this region. Temperature becomes very high. Hot and dry winds called **loo**, blow during the day.

Let's have fun :

1. People in all parts of our country drink delicious cool drinks called *Sharbat* made from fruits available in their regions. They are excellent thirst-quenchers and protect our bodies from the ill-effect of the harsh 'loo'. Have you tried 'Sharbat', made from raw mango, bel, lemon, tamarind, kokum, phalsa, watermelon and buttermilk made from curds; for example chhaachh, mattha, mori, chash, etc? Many make banana and mango milkshakes too.
2. After a hot summer, the first rains bring much joy. All our languages have melodious songs on 'rains'. They sound happy and bring cheer. Learn two songs on rains and sing them together. Write or collect five poems on rains. Ask your friends, neighbours and family members for names for rains and other seasons in different languages. For instance,
Varsha – Hindi Pous – Marathi
Barish – Urdu Borsha – Bengali



SOUTH WEST MONSOON SEASON OR RAINY SEASON

This season is marked by the onset and advance of monsoon. The winds blow from Arabian Sea and Bay of Bengal towards the land. They carry moisture with them. When these winds strike the mountain barriers, rainfall occurs.

SEASON OF RETREATING MONSOONS OR AUTUMN

Winds move back from the mainland to the Bay of Bengal. This is the season of the retreating monsoons. The southern parts of India, particularly Tamil Nadu and Andhra Pradesh receive rainfall in this season.

However, the **climate** is about the average weather condition, which have been measured *over many years*.

The climate of India has broadly been described as Monsoon type. **Monsoon** is taken from the Arabic word '**mausim**', which means seasons. Due to India's location in the tropical region, most of the rain is brought by monsoon winds. Agriculture in India is dependent on rains. Good monsoons mean adequate rain and a bountiful crop.

What would happen if monsoons were weak, or even worse, failed to occur one year? Tick (✓) the correct answer.

- Crop will be-
 affected/not affected
- The level of the water in a well will-
 come-up/go-down
- Summer will be-
 longer/shorter



Let's Do

On a map of India, locate the places mentioned in the paragraph.

The climate of a place is affected by its **location**, **altitude**, **distance from the sea**, and **relief**. Therefore, we experience regional differences in the climate of India. *Jaisalmer* and *Bikaner* in the desert of Rajasthan are *very hot*, while *Drass* and *Kargil* in Jammu and Kashmir are *freezing cold*. Coastal places like *Mumbai* and *Kolkata* experience *moderate climate*. They are neither too hot nor too cold. Being on the coast, these places are *very humid*. *Mawsynram* in *Meghalaya* receives the *world's highest rainfall*, while in a particular year it might not rain at all in *Jaisalmer* in Rajasthan.

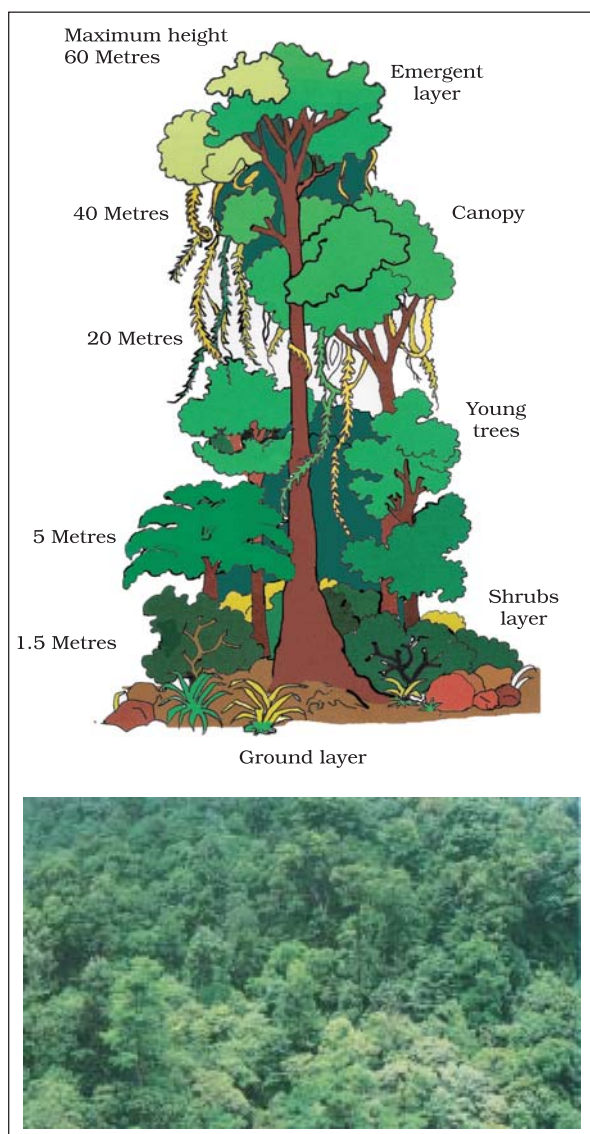


Figure 8.1 : Tropical Rain Forests

NATURAL VEGETATION

We see a variety of plant life in our surroundings. How nice it is to play in a field with green grasses. There are also small plants called bushes and shrubs like cactus and flowering plants etc. Besides there are many tall trees some with many branches and leaves like neem, mango or some which stand with few leaves such as palm. The grasses, shrubs and trees, which grow on their own without interference or help from human beings are called natural vegetation. Do you wonder how these differ from each other. Different types of natural vegetation are dependent on different climatic conditions, among which the amount of rainfall is very important.

Due to varied climatic conditions, India has a wide range of natural vegetation. Vegetation of India can be divided into five types – Tropical evergreen forest, Tropical deciduous forest, Thorny bushes, Mountain vegetation and Mangrove forests.

TROPICAL RAIN FOREST

Tropical Rain Forests occur in the areas which receive heavy rainfall. They are so dense that sunlight doesn't reach the ground. Many species of trees are found in these forests, which shed their leaves at different times of the

year. As a result, they always appear green and are called evergreen forest look at the Figure 8.1. Important trees found in these forests are *mahogany*, *ebony* and *rosewood*. Andaman and Nicobar Islands, parts of North-Eastern states and a narrow strip of the Western slope of the Western Ghats are home of these forests.

TROPICAL DECIDUOUS FORESTS

In a large part of our country we have this type of forest. These forests are also called monsoon forests. They are less dense. They *shed* their leaves at a particular time of the year. Important trees of these forests are *sal*, *teak*, *peepal*, *neem* and *shisham*. They are found in Madhya Pradesh, Uttar Pradesh, Bihar, Jharkhand, Chhattisgarh, Orissa, and in parts of Maharashtra.

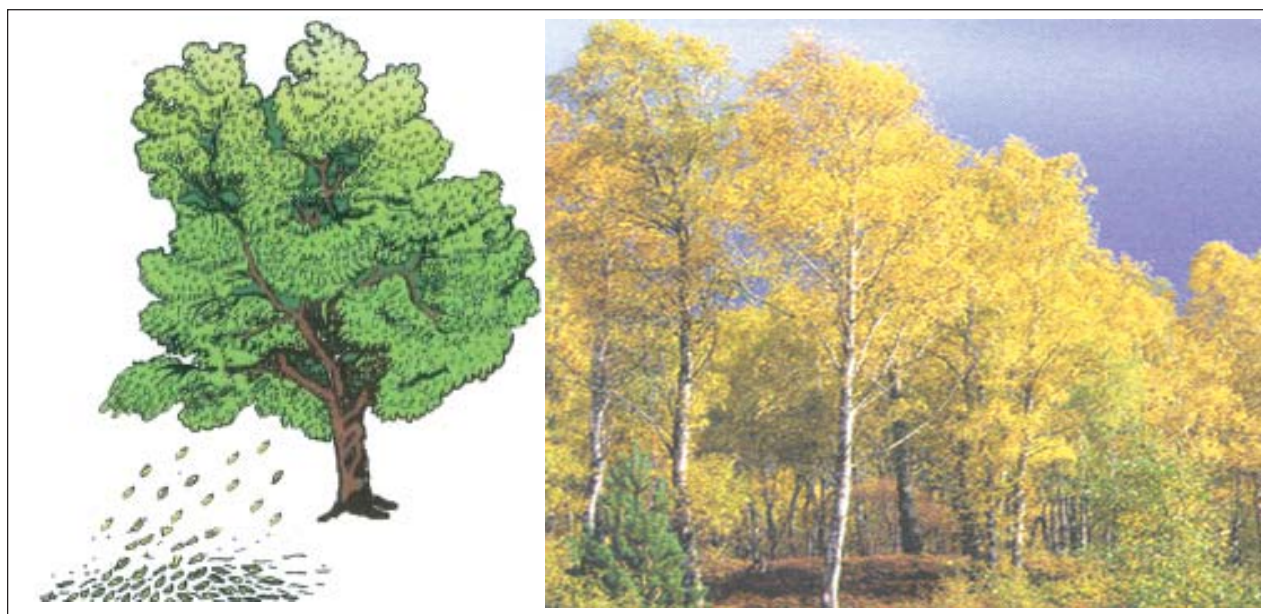


Figure 8.2 : Tropical Deciduous Forests

THORNY BUSHES

This type of vegetation is found in dry areas of the country. The leaves are in the form of spines to reduce the loss of water. *Cactus*, *khair*, *babool*, *keekar* are important and are found in the states of Rajasthan, Punjab, Haryana, Eastern slopes of Western Ghats and Gujarat.

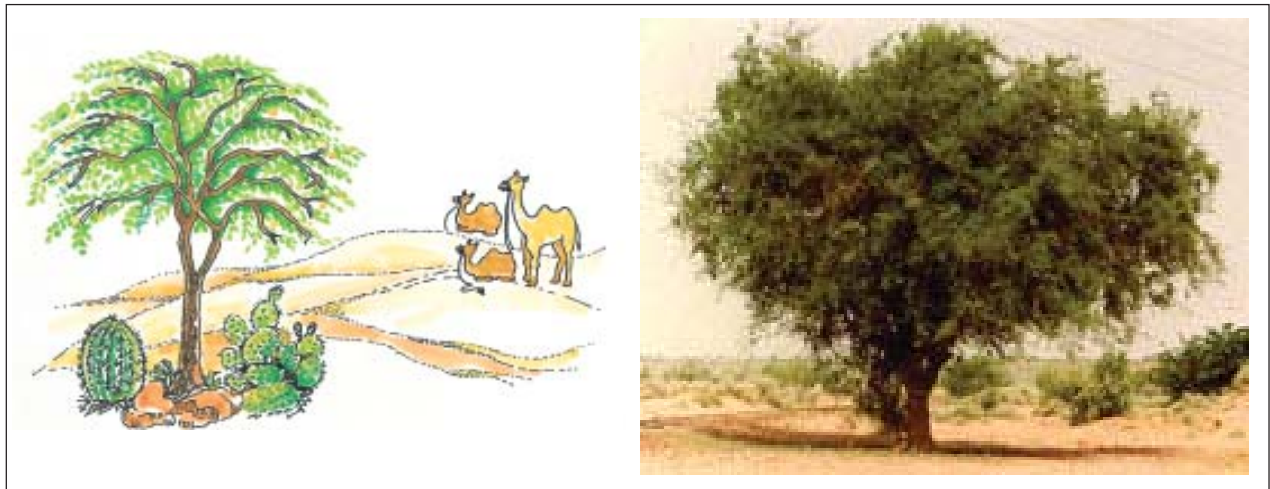


Figure 8.3 : Thorny Bushes

MOUNTAIN VEGETATION

A wide range of species is found in the mountains according to the variation in height. With increase in height, the temperature falls. At a height between 1500

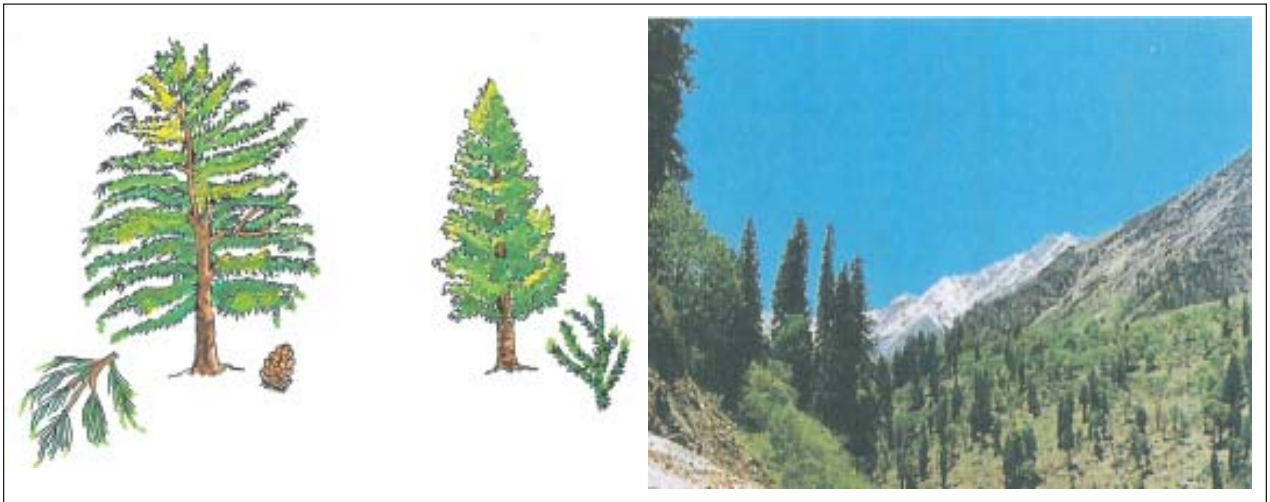


Figure 8.4 : Mountain Vegetation



Figure 8.5 : Mangrove Vegetation

metres and 2500 metres most of the trees are conical in shape. These trees are called coniferous trees. *Chir*, *Pine* and *Deodar* are important trees of these forests.

MANGROVE FORESTS

These forests can survive in *saline water*. They are found mainly in *Sunderbans* in

West Bengal and in the *Andaman and Nicobar Islands*. *Sundari* is a well-known species of trees in mangrove forests after which *Sunderbans* have been named.

WHY ARE FORESTS NECESSARY?

Forests are very useful for us. They perform various functions. Plants release oxygen that we breathe and absorb *carbon dioxide*. The roots of the plants bind the soil; thus, they control soil erosion.

Forests provide us with timber for furniture, fuel wood, fodder, medicinal plants and herbs, lac, honey, gum, etc.

Forests are the natural habitat of wild life.

Natural vegetation has been destroyed to a large extent because of the reckless cutting of trees. We should plant more trees and protect the existing ones and make people aware of the importance of trees. We can have special programmes like *Van Mahotsav* to involve more people in making our earth green.

Leela's parents planted a sapling of "neem" to celebrate her birth. On each birthday, a different sapling was planted. It was watered regularly and protected from severe heat, cold and animals. Children took care not to harm it. When Leela was 20, twenty-one beautiful trees, stood in and around her house. Birds built their nests on them, flowers bloomed, butterflies fluttered around them, children enjoyed their fruits, swung on their branches and played in their shade.

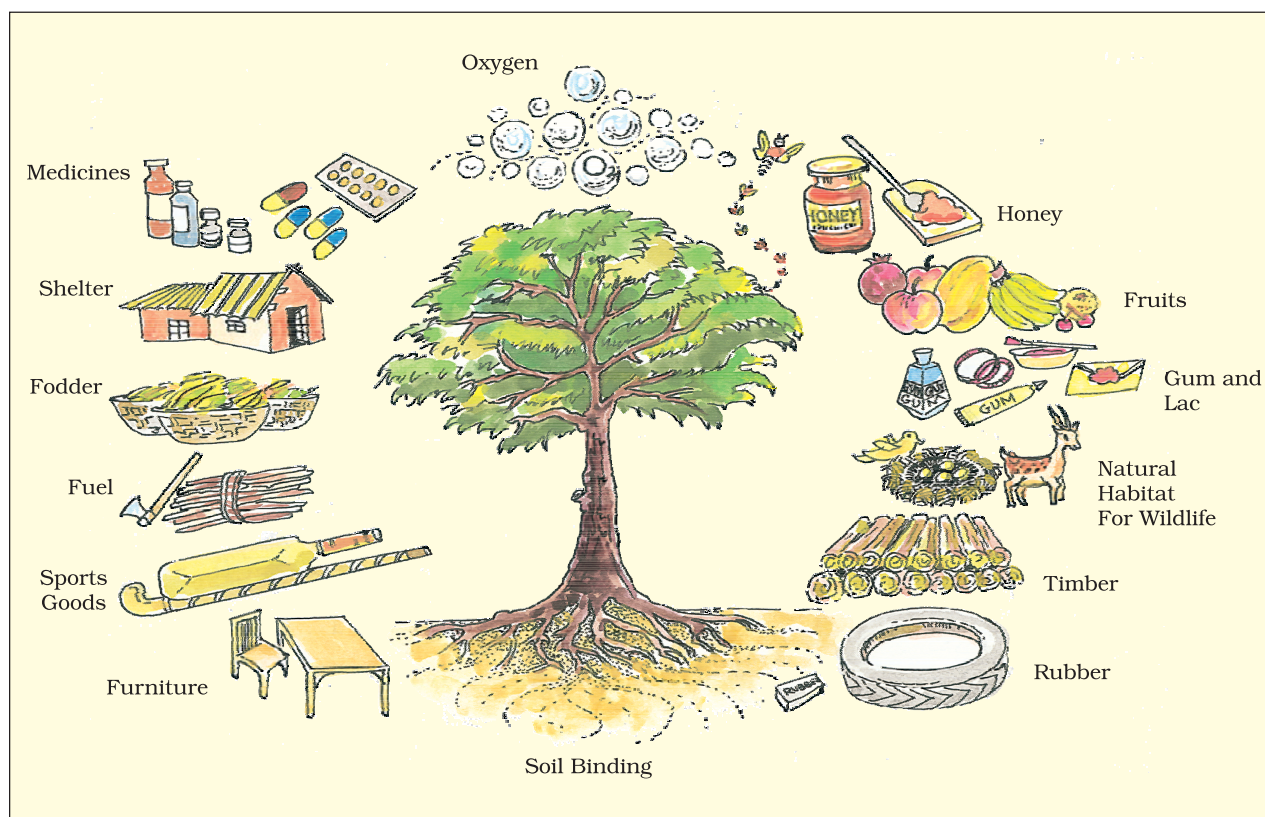


Figure 8.6 : Uses of Forests

WILD LIFE

Forests are home to a variety of wild life. There are thousands of species of animals and a large variety of reptiles, amphibians, mammals, birds, insects and worms which dwell in the forest.



Figure 8.7 : Wildlife

The tiger is our **national animal**. It is found in various parts of the country. *Gir* forest in Gujarat is the home of Asiatic lions. Elephants and one-horned rhinoceroses roam in the forests of Assam. Elephants are also found in Kerala and Karnataka. Camels and wild asses are found in the the Great Indian desert and the Rann of Kuchchh. Wild goats, snow leopards, bears, etc. are found in the Himalayan region. Besides these, many other animals are found in our country such as monkey, wolf, jackal, nilgai, cheetal, etc.

India is equally rich in bird life. The peacock is our **national bird**. Other common birds are parrots, pigeons, mynah, geese, bulbul and ducks. There are several bird sanctuaries which have been created to give birds their natural habitat. These provide the birds protection from hunters. Can you name five birds that are commonly found in your area?

There are several hundreds of species of snakes found in India. Cobras and kraits are important among them.

Due to cutting of forests and hunting, several species of wildlife of India are declining rapidly. Many species have already become extinct.

In order to protect them many national parks, sanctuaries and biosphere reserves have been set up. The Government has also started *Project Tiger* and *Project Elephant* to protect these animals. Can you name some wildlife sanctuaries of India and locate them on a map?

You can also contribute in conserving wildlife. You can refuse to buy things made from parts of the bodies of animals such as their bones, horns, fur, skins, and feathers. Every year we observe wildlife week in the first week of October, to create awareness of conserving the habitats of the animal kingdom.

Largescale poaching alleged in Simlipal reserve

By Arun Kumar Das/TNN

New Delhi: Yet another tiger sanctuary appears headed the Sariska way. Though officially there are 101 tigers in Orissa's Simlipal reserve, sightings have dropped to five this year, raising fears of largescale poaching in the state's largest tiger sanctuary.

Not only are fewer tigers visible, villagers have also stopped complaining about cattle kills by the jungle cats. There are 64 villages with 4 lakh residents in the areas surrounding Simlipal and 12,000 people live inside the sanctuary area.

Apart from tigers, the 2,750-sq-km sanctuary is home to 127 leopards, 665 elephants and hundreds of bison, barking deer, sambar and wild boars. Park records say there were 13 tiger sightings in 2003. The figure dropped to 7 in 2004.

Jitendra Kumar, district forest officer, Baripada, who is also in-charge of Simlipal, insists low sightings don't translate to fewer tigers.

"Because it is a different terrain here. There are seven rivers passing through the forest and about 500 water bodies and falls in the forest. There is no shortage of food or water in the forest, so tigers don't go for human or cattle killings," he said.

Visitors, Kumar says, don't have the patience to wait for tigers as they are always in hurry.

"Unlike other sanctuaries, there is no fixed tiger spot here. So, one has to wait patiently in different places in the core areas. So far, we have not come across any evidence of poaching in the forest."

Admitting the uncertainty, Simlipal...



IS ORISSA'S SIMLIPAL RESERVE HEADING THE SARISKA WAY?

partment official said lack of cattle kills was intriguing and hoped new sophisticated census methods would be more accurate.

In the case of Sariska and Ranthambore, two of the more important tiger reserves in India, previous census reports were either found to be exaggerated or completely wrong. Bhowaji Mohanty of the Wildlife Society of Orissa, which monitors the Simlipal tiger population, says claims of the authorities aren't valid and circumstantial evidence suggests otherwise.

"Unlike in elephant poaching, where the poacher leaves behind the carcass, the tiger poacher..."

"Because right from its nails, eyes, toes, teeth, it's thing is sold. There were 5 of leopards and tiger in the last four months in the park. There were no poaching cases were reported."

According to Wildlife T dia vice-chairman Ashu, who has moved a PIL in the Court challenging official tiger figures, "The number is not more than 30 as per matation. It is true that they of water bodies in the and it is also a dense forest..."

Villages in tiger reserves must be relocated

Q How did the PM react to the suggestions of the task force?

A The PM reacted in a positive manner. We had made seven recommendations in our report and the PM accepted all of them. Which recommendation needs to be implemented most?



SUNITA NARAIN
Chairman, Tiger Task Force
On the panel's meeting with...

portant, the plan is relocation of villages within a year

New Delhi: India's seriousness in saving its wildlife will be judged internationally by how quickly it can put in place one special agency to act against poachers. It promised to do so six years ago, it still hasn't and this has created a situation which criminals are exploiting.

The message coming out here, there is no specialised agency to act against poachers. The message coming out here, there is no specialised agency to act against poachers. The message coming out here, there is no specialised agency to act against poachers.

doesn't threaten their survival. China has been trying to communicate information to India on its activities but there is no one agency here which can act on it. agency here which can act on it. agency here which can act on it.

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Will Centre, wildlife lovers finally kiss and make up?

By Chandrika Mago/TNN

New Delhi: The country's new di-

the function. This certainly doesn't mean the two sides have kissed and made up.

More Sariskas in the making

By Chandrika Mago/TNN

New Delhi: There could be many more Sariskas in the making - and this is of field. If tourism could be tigers laying away from Rajasthan's famous Ranthambore reserve, "extremist" engaged disturbances" have taken an unaccounted toll on five other reserves.

India is supposed to have about 3,300 tigers but nobody believes the figure, just as nobody is quite sure of the scale of poaching. Over the past year or so, Project Tiger officials have been mapping traditional tiger ranges across the country to detail the changes and strategies accordingly. This is an attempt to put management on a scientific footing - untested acknowledgment that it hasn't been so far.

As the PM will hear at an upcoming National Board for Wildlife meeting, the traditional range has been mapped on the basis of literature references. Then, they have gone down in the table-ka level for actual sightings. They find the habitat is fragmented but the Terai, Western Ghats and central India are still not too bad."



ter of all sightings in and outside reserves. States, however, are the decisive factor in every reserve.

When the next tiger census begins November, the ministry plans to...

Gopal says the tourist crush at Ranthambore has pushed out tigers to villages nearby and animals have been sighted close to the Chambal ravines. At least one person's death has been reported. In the Valmiki reserve, a disputed area, no one has been sighted for nearly 10-15 months just washed out in pre-monsoon. Army patrolling has reportedly been sought there.

Other reserves in the problem list are Indravati in Chhattisgarh, Manas in Assam, Nagarjuna Sagar in Andhra Pradesh and Palamu in Jharkhand. "Some reserves are so disturbed you can forget about the tigers," says additional DG Wildlife P S Kaul.

Law and order is just a piece of the puzzle. A deadly cocktail has been brewing in reserves as the terrorising tiger loses its range, a human trample ground in dis-

its effort to protect poachers on the ill-armed men who are supposed to be a protective co-no leaving force.

There is between officials and tigers in the forest, who are best with the locals, who are best with the movements of both and poachers.

Big cats vanishing from Uttarakhand

By Jackson Chopra/This

Delhi: Dun: Uttarakhand's home to some of India's most famous tiger reserves, but sightings have been dropping steadily since the state was carved after the state re-

Wildlife Census 2002 report on Thursday. The report shows that the state had lost 10 tigers over the last four years - more than two tigers every year. While there were 221 tigers in 2001 and 240 in 2002, the number has further reduced to 140 in 2003. The Corbett Tiger Reserve (CTR) has tumbled from 140 to 100 compared to 140 in 2000. Over 100 tigers have been sighted in the last four years. In the Bhami tiger park, the number of tigers has remained the same since 2001 (20 tigers). How many are now 30 tigers compared to 30 in 2001. The number of tigers in the Bhami tiger park has remained the same since 2001 (20 tigers). How many are now 30 tigers compared to 30 in 2001.

The Door to the Himalayas, a tiger reserve in the state, has a tiger population of 100. The number of tigers in the Bhami tiger park has remained the same since 2001 (20 tigers). How many are now 30 tigers compared to 30 in 2001.

Cramped in Gir, lions pine for grassland



By Hemant Kumar/This

Cramped: In Gir, lions pine for grassland. The Gir Forest National Park in Gujarat is home to the last remaining wild lions in India. The park is a 1,412-sq-km area of dry deciduous forest. The lions are kept in a 1,412-sq-km area of dry deciduous forest. The lions are kept in a 1,412-sq-km area of dry deciduous forest.

Task force begins two-day meet

New Delhi: The Prime Minister's task force on tiger conservation is meeting over two days in the Capital, consulting activists, scientists, and government officials.

On Wednesday, the members sat for 9 am to 5 pm, hearing out the international conservationists on CITES as well as Indian through a group discussion which saw some of the last issues come out in the open. Thursday is for day for scientists - with a greater view counts and how to



- Why do poachers kill tigers?
- What will happen if tigers vanish from our forests?
- Have you ever visited any tiger reserves or a zoo where tigers are kept?

Migratory Birds

Some birds migrate to our country in the winter season every year such as Pelican, Siberian Crane, Stork, Flamingo, Pintail Duck, Curlew. Siberian Cranes migrate from Siberia; they arrive in December and stay till early march.



Stork – a migratory bird

EXERCISES

1. Answer the following questions briefly.

- (a) Which winds bring rainfall in India? Why is it so important?
- (b) Name the different seasons in India.
- (c) What is natural vegetation?
- (d) Name the different types of vegetation found in India.
- (e) What is the difference between evergreen forest and deciduous forest?
- (f) Why is tropical rainforest also called evergreen forest?

2. Tick the correct answers.

- (a) The world's highest rainfall occurs in
 - (i) Mumbai
 - (ii) Asansol
 - (iii) Mawsynram
- (b) Mangrove forests can thrive in
 - (i) saline water
 - (ii) fresh water
 - (iii) polluted water
- (c) Mahogany and rosewood trees are found in
 - (i) mangrove forests
 - (ii) tropical deciduous forests
 - (iii) tropical evergreen forests
- (d) Wild goat and snow leopards are found in
 - (i) Himalayan region
 - (ii) Peninsular region
 - (iii) Gir forests

- (e) During the south west monsoon period, the moisture laden winds blow from
- (i) land to sea (ii) sea to land (iii) plateau to plains

3. Fill in the blanks.

- (a) Hot and dry winds known as _____ blow during the day in the summers.
- (b) The states of Andhra Pradesh and Tamil Nadu receive a great amount of rainfall during the season of _____.
- (c) _____ forest in Gujarat is the home of _____.
- (d) _____ is a well-known species of mangrove forests.
- (e) _____ are also called monsoon forests.



FOR FUN

1. Make a list of trees in your surroundings and collect the pictures of plants, animals and birds and paste them in your copy.
2. Plant a sapling near your home and nurture it and write down the changes you observe for a few months.
3. Does any migratory bird come in your locality? Try to identify that. Be watchful in winter season.
4. Visit a zoo in your city or visit a nearby forest or sanctuary with your elders. Watch various types of wildlife there.

